

Finding the Right Process for Flow Chemistry - A Collaborative Workshop Guide

Why We're Doing This

At Flownetics, we believe every organization has the potential to benefit from flow chemistry, it's just about finding the right starting point.

When the right process meets the right strategy, the results can be game-changing: faster development, cleaner operations, safer scale-up, and reduced costs.

This workshop is designed to work with your team to uncover those **high-impact opportunities for flow implementation** and chart a clear, practical path from concept to commercialization.

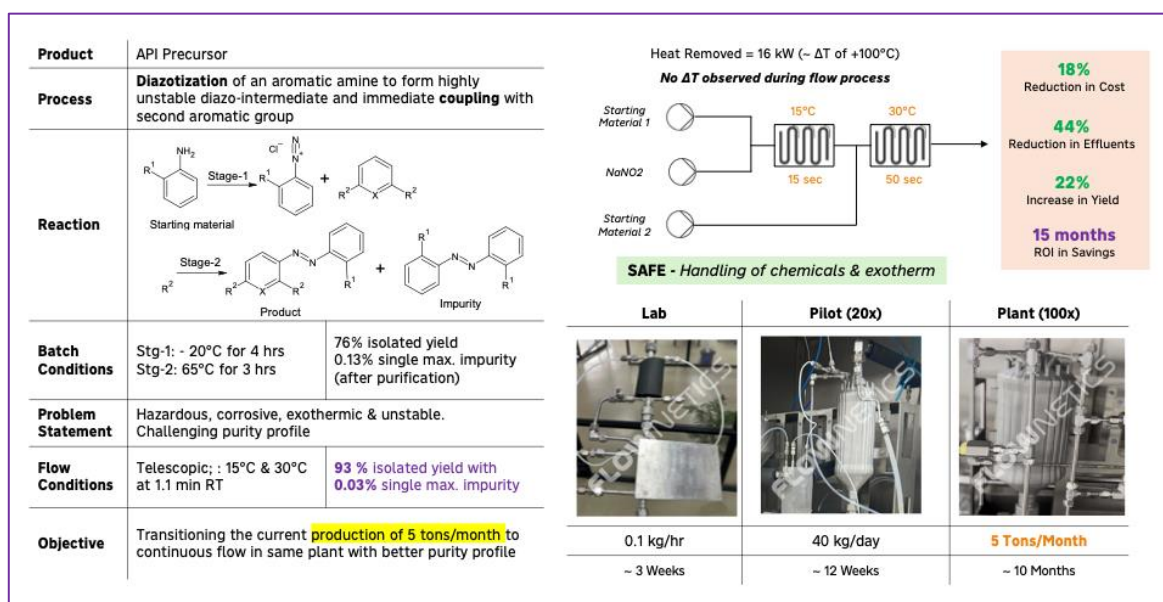
Step 1: What Are You Looking to Achieve?

Each company's motivation for adopting flow is different. Clarifying this early ensures we target projects that deliver measurable value quickly.

Are you looking for:

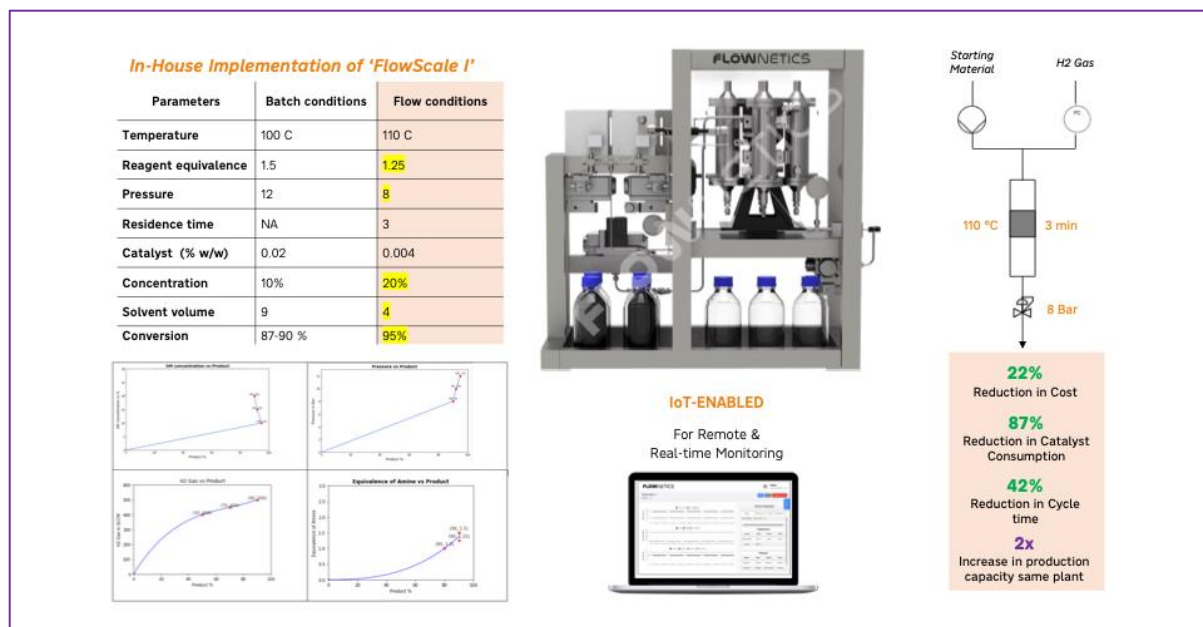
- Quick success stories?**

High-value or niche intermediates where flow can deliver immediate ROI, improve yields, or solve a critical bottleneck.



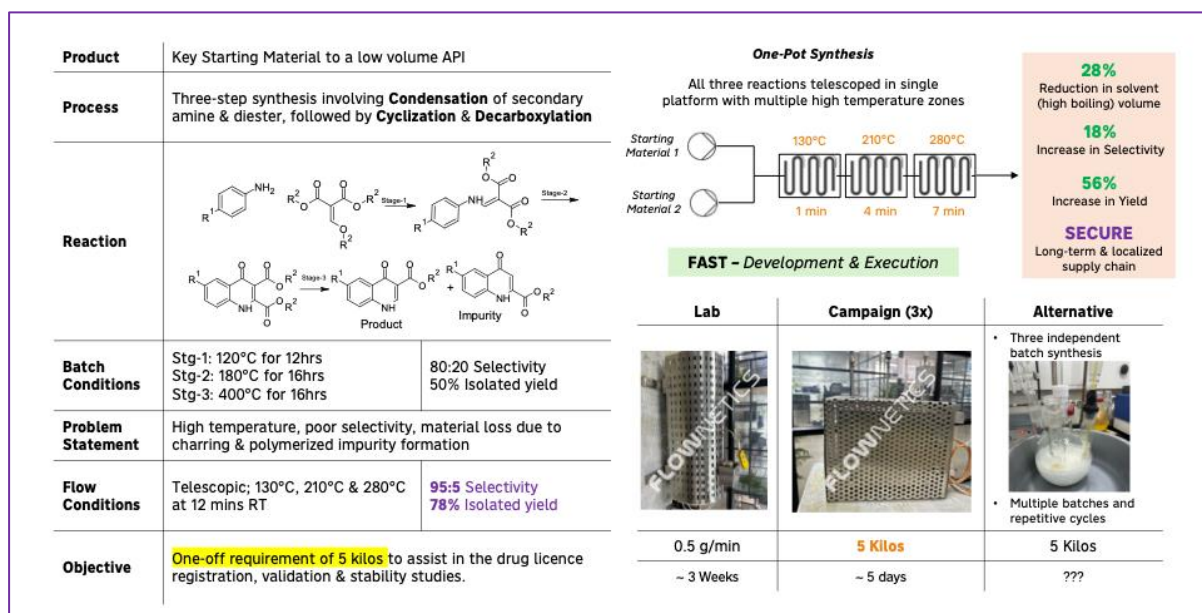
- Long-term development projects?**

Complex, multi-step reactions, new molecule routes, expansions or new product launches where process intensification can redefine your overall manufacturing strategy.



- Strategic or competitive benefits?**

Products tied to market access, sustainability targets, or supply chain independence where flow offers an edge.



- Alignment with ESG or CSR goals?**

Projects that reduce waste, energy, or solvent footprint while improving safety and compliance metrics.

Step 2: Identifying the Right Candidates

When assessing your portfolio, consider processes where batch limitations are most visible.

Here are areas where Flownetics has consistently delivered impact:

- Handling **hazardous or high-pressure reactions** more safely
- Solving **repeatability and consistency issues** in scale-up
- Improving **yield and conversion efficiency**
- **Increasing capacity** without expanding physical infrastructure
- Enabling **back-integration** of key intermediates or starting materials
- Reducing **energy and solvent consumption**
- Overcoming **scale-up bottlenecks** or validation delays
- Helping **late-stage or commercial products** meet tighter specifications

If you can relate to any of these, it's a strong signal that the process may benefit from a flow-based approach. **Understanding these implications are very important because the math doesn't always add up.**

Combining Our Strengths

Every organization brings deep process knowledge, production realities, and strategic priorities. At Flownetics, we complement that with the ability to handle a wide spectrum of chemistries from straightforward transformations to complex, multi-stage and high-risk operations. By combining both sides, we can quickly identify which categories of reactions are most likely to benefit from flow and how to make those transitions practical.

Core Strengths & Quick Turnaround

These are areas where we have deep experience and can move rapidly from feasibility to scale-up.

1. Fast & Exothermic Chemistry — nitrations, lithiations, diazotizations, high-energy substitutions
2. Heterogeneous Catalysis & Hydrogenations — reductive aminations, nitro reductions, gas-liquid-solid systems

Tailored Development (Longer Timelines)

These require custom engineering, mixing design, or extended process development but are well within our capability.

1. Slurry-Phase & Solid-Handling Reactions — crystallizing, viscous or multi-phase systems
2. High T/P & Solvent-Free Processes — intensified cyclizations, condensations, decarboxylations

Technologies Under Active Development

Not yet broadly available but progressing through internal R&D.

1. Halogenation using HF

2. Vapour-Phase Chemistry

Currently Inaccessible

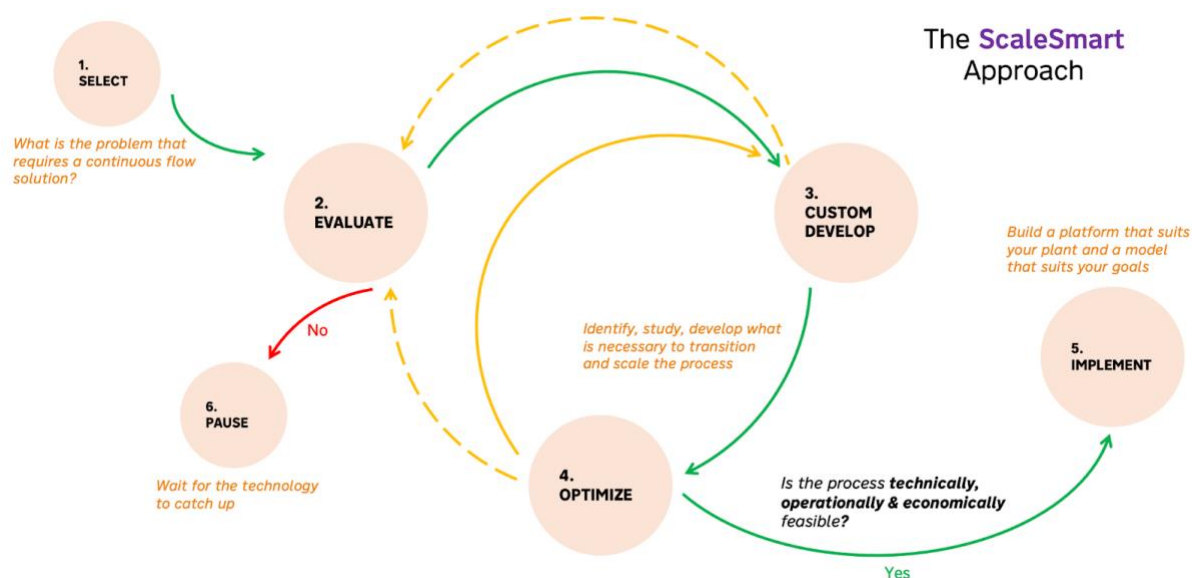
Outside our present platform scope but tracked for future expansion.

1. Photochemistry / Electrochemistry
2. Halogenation with HBr

Outcome: By mapping your chemistry across these capability levels, we can identify what can be accelerated now, what requires deliberate development, and what may become viable as new technologies mature.

Step 3: Making the Transition Practical

Flow chemistry success depends on more than just the chemistry, it requires the right balance of technical, operational, and commercial readiness. At Flownetics, we don't rush into manufacturing; we make every project using the **ScaleSmart Approach**. This ensures every process is robust, repeatable, and ready to deliver the impact promised by flow chemistry.



In our experience, the most successful projects have:

- A **moderate-to-high product value** that justifies transition costs and shared savings.
- A **cross-functional team** with strong collaboration between chemistry, engineering, and management.
- **Accessible process data**, so we can make early decisions based on facts, not assumptions.
- A shared understanding that **phase-wise scale-up** is the fastest, safest path to commercialization.

What Happens Next

In subsequent discussions, we'll:

1. Review 4 - 5 candidate products or intermediates from your portfolio.
2. Discuss their technical and business fit for flow implementation.
3. Classify them as:
 - Quick wins
 - Long-term developments
 - Strategic initiatives
4. Co-develop a roadmap for feasibility, development and possible commercialization under our **Factory-as-a-Service (FaaS)** model.

Closing Thought

The goal of this workshop is not to sell a solution but to **co-create a success story** - one that balances innovation with practicality.

Together, we can identify where flow truly adds value and build a scalable, sustainable pathway to bring those benefits to market faster.